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Exploring Gene–Environment Interactions: Socioregional Moderation of Alcohol Use

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Examples of gene–environment interaction in human behavioral data are relatively rare; those that exist have used simple, dichotomous measures of the environment. The authors describe a model that allows for the specification of more continuous, more realistic variations in environments as moderators of genetic and environmental influences on behavior. Using data from a population-based Finnish twin study, the authors document strong moderating effects of socioregional environments on genetic and environmental influences on adolescent alcohol use, with nearly a five-fold difference in the magnitude of genetic effects between environmental extremes. The incorporation of specific environmental measures into genetically informative designs should prove to be a powerful method for better understanding the nature of gene–environment interaction and its contribution to the etiology of behavioral variation.

That behavior results from a complex interplay of genetic and environmental influences is widely accepted, yet gene–environment interaction has proven difficult to document in human data. However, large-sample twin studies, advances in biometrical modeling, and the inclusion of environmental measures in genetically informative designs now enable researchers to test whether environmental variation alters the impact of dispositional genes for behavior. Early documentation of that effect was found in data from the Australian twin register: Marital status moderated the relative importance of genetic effects on alcohol consumption (Heath, Jardine, & Martin, 1989) and on depression symptoms (Heath, Eaves, & Martin, 1998) in female subjects. Having a marriagelike relationship reduced the impact of genetic influences on drinking: Among the younger sample of twins, genetic liability accounted for only half as much variance in drinking among married women (31%) as among unmarried women (60%). Similarly, among adult twins (>30 years), genetic effects accounted for less than 60% of the variance in married respondents but greater than 76% in unmarried respondents (Heath et al., 1989). A marriagelike relationship also reduced the influence of genetic liability to depression symptoms: Genetic factors accounted for 29% of the variance in depression scores among married women but for 42% of the variance in young unmarried women and 51% of the variance in older unmarried women (Heath et al., 1998).

Life events also moderate the impact of factors influencing depression in women (Kendler, Neale, Heath, Kessler, & Eaves, 1991). Genetic and/or shared environmental influences were significantly more important in influencing depression in high-stress than low-stress environments, as defined by a median split on a life-event inventory, although there was insufficient power to determine whether that moderating influence was on genetic or environmental effects.

More recently, religiosity has been found to have a moderating effect: Religiosity moderated disinhibition scores, as measured by the Sensation Seeking Scale, in a sample of Dutch men (Boomsma, de Geus, van Baal, & Koopmans, 1999). In subjects without a religious upbringing, genetic effects played a large role, dwarfing the influence of shared environment. However, in individuals with a religious upbringing, there was no evidence of genetic influence, whereas shared environment played a large role. A parallel moderating influence of religiosity has been reported for alcohol use among women (Koopmans, Slutske, van Baal, & Boomsma, 1999).

In adolescent Finnish twin data, we have reported that regional residency exhibits moderating effects on influences on religiosity (Winter, Kaprio, Viken, Karvonen, & Rose, 1999) and alcohol use

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(Rose, Dick, Viken, & Kaprio, 2001; Rose, Kaprio, Winter, Koskenvuo, & Viken, 1999). In adolescent women's religiosity, unique environmental influences were more important among those living in the urban and more secular south; for boys, however, there were increased genetic effects and decreased common environmental effects on religiosity in the urban south (Winter et al., 1999). The importance of parental and sibling influences on adolescent abstinence also differed in urban and rural settings (Rose et al., 1999). Cooperative sibling effects were more evident among twins living in Greater Helsinki, where abstinence was relatively rare, than among twins living in Northern Finland. Conversely, the buffering effect of parental abstinence on their adolescents' abstinence was more evident in Northern Finland, where abstinence was more prevalent. The effect of parents was masked in the urban areas around Finland, where abstinence was less prevalent and sibling and peer influences assumed a greater role.

These initial analyses of moderation of sibling and parental influences on abstinence by regional residency led us to more sophisticated modeling of the effect of urban/rural residency on influences on drinking frequency (Rose et al., 2001). All twins were classified as residing in urban or rural environments at age 16 on the basis of Finnish government statistical records. Although we found no main effect of urban/rural residency on frequency of alcohol use, a significant gene–environment interaction emerged: Genetic factors exerted a larger role in urban settings than in rural settings from age 16 through age 18.5, whereas common environmental factors assumed greater importance in rural settings.

Together, the studies cited above reinforce the idea that environments moderate the impact of genetic and environmental influences across many dimensions of behavior. The moderating role of religiosity and regional residency suggests that environments that are less controlling and provide a greater range of opportunities to express individual differences allow for greater expression of dispositional genes. The findings by Heath et al. (1989, 1998) and Kendler et al. (1991) suggest that "protective" environments, characterized by marriagelike relationships or low stress levels, can reduce the impact of genetic predispositions to various clinical problems. However, not all studies of gene–environment interaction have reached that conclusion: The National Longitudinal Study of Adolescent Health found that self-reported family warmth averaged for all children in the same school moderated the influence of genetic and environmental factors impacting adolescent aggression. Heritability increased and common environmental effects decreased with a greater level of family warmth (Rowe, Almeida, & Jacobson, 1999). The authors hypothesized that in more supportive, benign environments, genetic predispositions were required for the development of aggression; however, in more undesirable, less supportive environments, external environmental influences (such as deviant peers) became more influential in the manifestation of aggression. This discrepancy suggests that gene–environment interplay varies in different disorders, and perhaps in different age groups, as Heath and Kendler studied adults and Rowe and colleagues studied adolescents.

These studies have been important steps in documenting the presence of gene–environment interaction, but they do not begin to capture the complexity of environmental effects. All of these studies have used dichotomous measures of the environment. In some cases, the environmental measures studied lended themselves to dichotomization, as in the study of marital status. However, other environmental influences, such as family warmth and stress levels, may be more accurately captured with more continuous measures. When the relationship between the moderator and the impact of genetic and environmental influences is linear in nature, more complete, more realistic characterization of the environment should enhance our understanding of that interaction.

We sought to investigate the nature of gene–environment interaction by using more continuous measures of the environment, by following up on our previously reported finding that regional residency moderates genetic and environmental influences on alcohol use. Urban/rural residency itself is not informative as to the underlying process. What about urban settings allows for greater expression of genetic effects? Why do common environmental influences exert a larger influence in rural settings?

To explore these questions, we developed a model in Mx (Neale, Boker, Xie, & Maes, 1999) that allowed for moderating environments with multiple levels. We examined a series of socioregional variables we hypothesized to be more proximal to the interaction effect. Note that most studies examining the importance of group-level influences on adolescent alcohol use have focused on familial and peer influences, with relatively little attention paid to socioregional variables. This observation led one sociologist to regretfully note, "The classic sociological theories of drug behavior have given way to social-psychological frameworks in which the explanatory emphasis is on individual attributes rather than on broad social-structural factors" (Kandel, 1980). More recent reviews have reached a similar conclusion (Johnstone, 1994). Those studies that have examined socioregional variation in alcohol use have limited their reports largely to mean differences rather than incorporating this information into theoretical models of the development of alcohol use and abuse (Arvers & Choquet, 1999; Jenkins et al., 1998; Madianos, Gefou-Madianou, & Stefanis, 1995). For such reasons, we sought to more thoroughly explore the nature of gene–environment interaction and to address unanswered questions regarding the importance of socioregional levels of environmental influence. Finland offers a unique opportunity to explore such questions because the country is divided into municipalities: local government units that levy taxes and assume responsibility for education, health care, and social services. Finland's municipalities range from those in the metropolitan Helsinki area to sparsely populated communities in Finnish Lapland and the Åland Islands. A wealth of sociodemographic information, relevant to risk for adolescent use/abuse of alcohol, can be linked to each municipality. We linked such information to each twin pair's municipal residency code to identify risk-relevant characteristics about the community in which each twin pair lived, to study the influence of community-level factors on adolescent drinking outcomes.

In an attempt to better characterize the urban/rural interaction effect, we examined the percentage of young adults, ages 20–24, in the different municipalities, the percentage of migration into and out of different regions, and the per capita expenditure on alcohol in each region, relative to the mean amount spent in the entire country. Each of these three variables was chosen on theoretical grounds. We expected that there would be a larger proportion of young adults in urban settings, where youth tend to move on leaving home. This larger proportion of young adults, slightly older than our adolescent twins, would offer a broader range of

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potential role models for the twins. Thus, the urban environment would provide a wider range of potential friends and activities, allowing for greater expression of different genetic predispositions. Percentage migration into and out of the region was examined under the assumption that urban settings would have more mobility and that greater mobility in a local setting would create a more anonymous environment. Conversely, in a rural setting where there is more residential and generational stability, there may be more community monitoring, engendering more accountability for one's actions and more shared pressure to engage, or not engage, in certain behaviors. Once again, this would create an atmosphere allowing for greater expression of genetic effects in urban settings and greater importance of common environmental influences in rural settings. Finally, we anticipated that there would be proportionately more money spent on alcohol in urban settings, where retail outlets of the state alcohol monopoly are much more prevalent. This would create an environment in which there is greater accessibility and, therefore, increased opportunity to choose settings more accepting of adolescent drinking, allowing individuals to express a range of genetic predispositions.

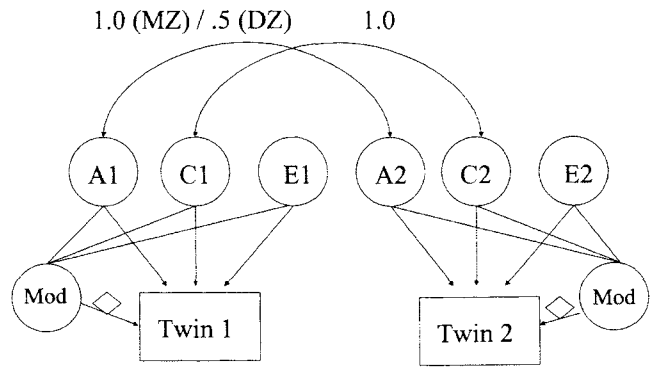

Figure 1. Moderator model. A1 and A2 represent genetic influences on Twin 1 and Twin 2, respectively. C1 and C2 represent common environmental influences, and E1 and E2 represent unique environmental influences. The definition variable, represented by a diamond, carries the value of the specified moderator (Mod) for each twin. For simplicity, means are not represented in the diagram but are included in the model when using raw data analysis.

Method

FinnTwin16 (FT16) is a population-based developmental twin study of health risk factors. It consists of five consecutive birth cohorts of twins born during 1975–1979. All twin pairs were ascertained through Finland's Central Population Registry (CPR), which permits exhaustive and unbiased ascertainment of all twins living and resident in the country through unique identifying numbers assigned at birth. Baseline questionnaires were mailed to all twins within 2 months of their 16th birthdays. Follow-up questionnaires were mailed to all respondent twins at ages 17 and 18.5. Response rate was approximately 90% across all waves of data collection. Further details on FT16 are reported elsewhere (Rose et al., 1999).

Socioregional variables of interest were obtained from *SuomiCD*, which contains a collection of Finnish government statistics, such as average income and population density, compiled for each of the country's 452 municipalities (Statistics Finland, Helsinki, http://www.stat.fi/tk/tp_db/suomicd.html). We obtained the twins' residential municipalities from the CPR at baseline and then linked this socioregional information in *SuomiCD* to the individual twins in FT16. Co-twins in all but two pairs were living in the same municipality; those two pairs were removed, so that all pairs in analyses reported here were concordant for municipal residential status at baseline. In relation to the urban/rural interaction, we examined (a) the percentage of young adults, ages 20–24, in each area; (b) the percentage of migration in and out of the area; and (c) the amount of money spent (sales) on alcohol in each of the areas relative to the overall mean in the country. The percentage of young adults varied from 2% to 9% across the different municipalities. Percentage of migration varied from 2% to 16%, and relative amount of money spent on alcohol ranged from 11% to 157% of the mean amount for the entire country. Because computational difficulty increased substantially with the number of moderator levels, we grouped sales into five approximately equal groups representing the lowest to highest quintiles of the distribution.

We first tested whether each variable correlated with urban/rural residency and then tested whether the variable also had a moderating influence on alcohol use that paralleled the urban/rural interaction, suggesting that the variable may, in part, be accounting for the effect. Pearsonian correlations between socioregional variables and urban/rural residency were computed using the statistical package SPSS. Moderation models (Figure 1) were developed and run using the raw maximum likelihood feature in Mx (Neale et al., 1999). Genetically informative models permit variance in a behavior to be partitioned into additive genetic influences, common environmental influences, and unique environmental influences. Genetic influences correlate 1.0 between monozygotic (MZ) twins, who share all of their genes identical-by-descent, and 0.5 between dizygotic (DZ) twins, who share (on average) 50% of their segregating genes, as do ordinary siblings. Common environmental effects, as defined in biometrical twin modeling, refer to all environmental influences that make siblings more similar to one another. By definition, these influences correlate 1.0 between both MZ and DZ twins. Unique environmental influences are uncorrelated between co-twins and have the effect of decreasing the covariance between siblings. Note that environmental effects are assigned as common or unique in biometrical twin studies on the basis of the effect that they produce in siblings: making them more similar or different. These terms do not designate whether the environment is objectively shared by siblings; thus, although siblings may share the experience of parental divorce, if the divorce has the effect of making the siblings more different from one another, the variance due to this environmental event will be attributed to unique environment. (For a complete review of this topic, see Turkheimer & Waldron, 2000.)

Mx allows the use of a special kind of variable in conjunction with raw maximum likelihood estimation. The value of a path can be fixed to a definition variable. Unlike the fixed paths in conventional analyses of covariance matrices, which apply the same value to all individuals in the sample, the path parameter fixed by a definition variable changes from subject to subject, depending on the value of the definition variable; essentially a separate model is fit to subjects with each level of the definition variable. In this way, the use of definition variables is similar to the use of multiple groups to represent different environments. However, the definition variable allows a more parsimonious representation of a range of environmental conditions and provides a model that is similar to other multilevel approaches. In our models, the different socioregional variables were used as definition variables; thus, the moderators differ between twin pairs but not within twin pairs. It would, of course, be possible to specify moderators that differ within twin pairs as well, for example, to study the effect of community factors on twins once they have moved away from home and are residing in different municipalities.

The full model allowed for the definition variable to moderate genetic, common environmental, and unique environmental influences. By dropping parameters, we tested the significance of the moderating effect on each of these respective influences. Differences between the log-likelihoods of nested models were compared with a chi-square distribution. When there was a significant chi-square ($p < .05$) for the difference in

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degrees of freedom between the models, the model with fewer degrees of freedom was adopted.

Drinking frequency was measured with a 9-point scale in a question that asked, "How often do you drink alcohol at all?" (1 = *daily*, 9 = *I don't drink*). We recoded these responses into drinking days per month and treated drinking frequency as a continuous variable. We then log-transformed the data to improve the normality of the distribution of drinking frequencies. Here, we report analyses of drinking frequency at age 18.5, as by this age, adolescents' drinking patterns are generally more stable than initial, experimental use earlier in adolescence. Only twin pairs that reported concordant drinking at age 18.5 were used in the analyses to study influences on frequency of alcohol use, without confounding frequency of use with other dimensions of substance use such as initiation and persistence, each of which have their own sources of genetic and environmental influences (Heath & Martin, 1993). Concordant drinking was reported by 1,580 same-sex twin pairs (462 female MZ pairs, 392 female DZ pairs, 327 male MZ pairs, and 399 male DZ pairs) at age 18.5.

Results

Both the percentage of young adults in different communities and the amount of money spent (sales) on alcohol in different neighborhoods correlated negligibly, albeit significantly, with frequency of alcohol use (young adults, $r = .04$, $p = .03$; sales, $r = .08$, $p < .001$). These main effects were regressed out of the frequency of drinking measure for moderation analyses, although there was virtually no change from the original modeling results to those obtained using the corrected scores, emphasizing how negligible were the main effects of these variables. Migration did not correlate significantly with frequency of alcohol use ($r = .03$, *ns*).

The percentage of young adults, percentage of migration, and alcohol sales all correlated significantly with urban/rural residency. There was a larger percentage of young adults in urban than rural settings ($r = .71$, $p < .001$). There was more migration in urban settings ($r = .78$, $p < .001$) and relatively more money was spent on alcohol in urban than rural environments ($r = .34$, $p < .001$). Accordingly, each variable was tested as a potential moderator.

Young Adults

Fit statistics for each of the models, using the percentage of young adults in the municipality as the moderator, are shown in Table 1. The moderating effect of young adults on unique environmental factors was not significant; however, moderating effects on genetic and common environmental effects could not be dropped without causing a significant decrease in fit of the model. Constraining the young men and women to be equal on raw parameter estimates caused a significant decrease in fit of the model. However, the standardized estimates of A (additive genetic effects), C (common environmental effects), and E (unique environmental effects) could be set equal, as could the standardized moderating effect, indicating that the gender difference was due to the significant difference in variance between young male and young female adults' drinking patterns at this age. As the percentage of young adults increased, the average change in heritability increased .07 between each of the eight environments; the average change in common environmental effects decreased .07 between environments. Estimates of the amount of variance attributed to A, C, and E effects for municipalities with the lowest and highest percentages of young adults are shown in Table 2. Note that these are the implied estimates for the extreme environments, derived from data from the whole sample. Note that not only did the standardized estimates evidence a change in the relative importance of genetic and environmental effects but raw estimates of variance for A and C also changed: As the percentage of young adults in an area increased, genetic variance increased and common environmental variance decreased. That indicates that the actual amount of variance attributable to genetic and environmental factors, and not just the relative influence attributed to each, was changing across the different environments. There were no notable changes in total variance between environments.

Table 1
Model-Fitting Results for Drinking Frequency With Young Adults Moderating

Model	−2lnL	*df*	Tested against	χ^2 *p* value
1. Men and women differ—A, C, and E moderator effects	5212.003	2282		
2. No E moderation	2512.530	2284	1	*ns*
3. No A or E moderation	5220.432	2286	2	<.05
4. No C or E moderation	5221.328	2286	2	<.05
5. Men and women equal—raw parameters	5233.752	2888	1	<.05
6. Men and women equal—standardized effects	5218.696	2888	1	*ns*

Note. A = additive genetic effects; C = common environmental effects; E = unique environmental effects. Dropping each of these effects results in a change of 2 degrees of freedom because pathways were estimated separately for young men and young women. −2lnL = log-likelihood.

Migration

Results from model fitting with migration as the moderating variable are presented in Table 3. As for young adults, the moderating effect on unique environmental factors could be dropped; however, the moderating effect of percentage of migration on the genetic and common environmental factors could not be dropped without causing a significant decrease in fit of the model. As the percentage of migration increased, genetic variance increased and shared environmental variance decreased. The model constraining young men and women to be equal caused a significant decrease in fit; however, this difference was again due to variance differences, as the standardized estimates and moderating effects could be set equal. The average change in heritability between each of the 15 environmental levels of migration was .03, and the average change in common environmental effects was −.03. Standardized estimates of genetic and environmental influences in environments with the lowest and highest migration rates are presented in Table 4.

Regional Alcohol Sales

Model-fitting results for regional alcohol sales are presented in Table 5. There was no significant moderating effect on unique environmental factors. Nor did the moderating effect of sales on genetic influences reach significance. However, the amount of money spent on alcohol in different communities did significantly moderate the influence of common environmental factors: As the

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Table 2
Standardized Estimates (With Confidence Intervals) From Environments With the Fewest and Most Young Adults

Environment	A	C	E
Fewest young adults	0.13	0.52	0.35
(95% CI)	(.01–.39)	(.22–.69)	(.24–.48)
Most young adults	0.63	0.00	0.37
(95% CI)	(.43–.71)	(.00–.14)	(.29–.48)

Note. A = additive genetic effects; C = common environmental effects; E = unique environmental effects; CI = confidence interval.

Table 4
Standardized Estimates (With Confidence Intervals) From Environments With the Least and Most Migration

Environment	A	C	E
Least migration	0.16	0.48	0.36
(95% CI)	(.02–.40)	(.23–.64)	(.27–.46)
Most migration	0.60	0.01	0.39
(95% CI)	(.41–.70)	(.00–.14)	(.30–.50)

Note. A = additive genetic effects; C = common environmental effects; E = unique environmental effects; CI = confidence interval.

amount of money spent on alcohol in a region increased, the impact of shared environment significantly decreased. Standardized estimates of genetic and environmental influences in different environments are presented in Table 6. Although the table indicates considerable change in both genetic and environmental effects, the reader is reminded that the significance of moderating effects is based on differences at the level of raw variances, whereas the parameter values presented in the tables reflect standardized effects. In the case of regional alcohol sales, the raw variance changed more substantially between environments for common environmental effects than for genetic effects, explaining why only moderation of common environmental effects reached significance.

Young men and women could not be equated without causing a significant decrease in fit of the model; however, for regional alcohol sales we were unable to determine the exact origin of this difference. The standardized estimates of A, C, and E could be constrained to be equal for men and women without causing a significant decrease in fit, or the average moderating effect from one environment to the next could be constrained to be equal in both sexes. However, forcing both the standardized estimates and moderating effect to be equal in men and women caused a significant decrease in fit. Because we were unable to determine the exact men/women difference, A, C, and E estimates are presented separately for each sex (Table 6).

Discussion

The relative proportion of young adults in a regional area, the frequency of migration in and out of a region, and the relative amount of money spent on alcohol in an area all moderated influences on frequency of alcohol use among drinking adolescents. Sales appears to have acted in a somewhat different manner than the young adults and migration moderators, as it significantly moderated only common environmental effects. In addition, the magnitude of the moderating effect was smaller for regional alcohol sales, with substantial overlap in the confidence intervals for estimates for different environments. The percentage of young adults and the amount of migration in different neighborhoods strongly moderated genetic and common environmental effects, with a four-fold difference or more in the magnitude of genetic/common environmental effects between areas with few young adults and little migration to areas with many young adults and higher migration. As hypothesized, these results suggest that communities characterized by more young adult role models and greater social mobility allow for increased expression of genetic dispositions that contribute to individual differences in adolescent drinking. Conversely, communities with fewer older peers and more social structure create opportunities in which common environmental effects, within families and within communities, assume greater importance.

Drinking frequencies reported by Finnish adolescents place them in the mid-range of those reported by age-matched Europeans from 25 countries and by adolescent Americans of the same age. For example, in mid-adolescence, Finns report drinking with

Table 3
Model-Fitting Results for Drinking Frequency With Migration Moderating

Model	−2lnL	*df*	Tested against	χ^2 *p* value
1. Men and women differ—A, C, and E moderator effects	5215.339	2282		
2. No E moderation	5216.150	2284	1	*ns*
3. No A or E moderation	5222.217	2286	2	<.05
4. No C or E moderation	5224.048	2286	2	<.05
5. Men and women equal—raw parameters	5238.971	2888	1	<.05
6. Men and women equal—standardized effects	5219.196	2888	1	*ns*

Note. A = additive genetic effects; C = common environmental effects; E = unique environmental effects; −2lnL = log-likelihood.

Table 5
Model-Fitting Results for Drinking Frequency With Regional Alcohol Sales Moderating

Model	−2lnL	*df*	Tested against	χ^2 *p* value
1. Men and women differ—A, C, and E moderator effects	5206.866	2282		
2. No E moderation	5210.582	2284	1	*ns*
3. No A or E moderation	5215.554	2286	2	*ns*
4. No C or E moderation	5225.087	2286	2	<.05
5. Men and women equal—raw parameters	5230.180	2888	1	<.05
6. Men and women equal—standardized A, C, E	5210.062	2885	1	*ns*
7. Men and women equal—moderation	5207.417	2885	1	*ns*
8. Men and women equal—both standardized effects	5226.182	2888	1	<.05

Note. A = additive genetic effects; C = common environmental effects; E = unique environmental effects; −2lnL = log-likelihood.

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Table 6
Standardized Estimates (With Confidence Intervals) From Environments With the Lowest and Highest Regional Alcohol Sales

Environment	A	C	E
Men			
Lowest sales	0.32	0.29	0.39
(95% CI)	(.05–.67)	(.00–.54)	(.28–.51)
Highest sales	0.49	0.14	0.38
(95% CI)	(.16–.72)	(.00–.42)	(.28–.50)
Women			
Lowest sales	0.25	0.27	0.48
(95% CI)	(.02–.58)	(.00–.49)	(.37–.60)
Highest sales	0.61	0.01	0.38
(95% CI)	(.33–.71)	(.00–.23)	(.29–.50)

Note. A = additive genetic effects; C = common environmental effects; E = unique environmental effects; CI = confidence interval.

much less frequency than do same-age adolescents from Ireland, Wales, Scotland, England, or Italy (Hibell et al., 1997); however, similar to older adults in Nordic cultures, drinking among Finnish adolescents is often high-density drinking to intoxication. Yet, the critical question for the generalizability of our results is not addressed by these similarities and differences in descriptive patterns of drinking across cultures. The question of generalizability is in whether or not estimates of genetic and environmental influences on adolescent alcohol use, from large and representative twin samples, are generally consistent across cultures. All available data suggest high consistency in comparisons of twin data from Finland with data from the Netherlands, Australia, and the United States (Heath & Martin, 1988; Koopmans & Boomsma, 1996; Maes et al., 1999; Rose et al., 2001). It is, however, quite plausible that environmental moderators of influences on alcohol use may be more culture specific. The results reported here may reflect recent changes in alcohol policy in Finland: Prior to 1995, when Finland joined the European Union, a state-run alcohol monopoly strictly regulated licensing and availability of alcohol. Although the state monopoly on retail sales of alcohol continued after Finland joined the European Union, alcohol became much more available through loosened regulations on licensing of restaurants and increased operating hours of bars. At the same time, public opinion became much more liberal, and it is likely that urban areas were the first to adopt these new ideas, creating less restrictive environments within which genetic predispositions would find greater expression than in more stable, rural communities. Thus, although the general implications of these gene–environment interaction analyses are universal, the specific moderator variables may differ from one society and time to another.

Our results also provide a potential explanation for heterogeneity among heritability estimates yielded from different samples. Heritability estimates are population specific and can vary from sample to sample. Although heritability estimates are generally consistent between studies using population-based sampling strategies or large twin registers, variation among estimates from past twin studies, which generally used smaller sample sizes, was commonplace and led some researchers to question the utility of heritability estimates. Our results suggest that some of this variation may be due to environmental differences, as our findings clearly demonstrate that the importance of genetic and environmental effects can vary because of environmental circumstances. With smaller, selected twin samples, environmental differences between samples would be expected, potentially leading to differences in the magnitude of importance of genetic and environmental effects. Population-based studies presumably are more representative of the range of potential environments, thereby minimizing the overall influence of environmental variation on estimates of variance and leading to more consistent estimates. However, populations that vary substantially on mean levels of important environmental moderators could still yield different heritability estimates.

Do the three socioregional variables tested here account for the original urban/rural interaction we set out to further explore? To address that question, we tested the significance of the original urban/rural moderator, after regressing out the effects of the more specific moderators analyzed here. After regressing out the effect of young adults, migration, and sales, urban/rural residency did not have a significant moderating effect on adolescent alcohol use. This suggests that the moderators tested here are more proximal to the underlying psychological processes involved in the gene–environment interaction suggested by our original analysis of urban/rural residency (Rose et al., 2001). Understanding the processes involved in gene–environment interaction should enable us to use that information to inform intervention efforts. Exploring more detailed and theoretically proximal aspects of the environment provides more direct paths for designing effective prevention and intervention programs.

There is some question regarding the independence of each of these socioregional moderators. Regional alcohol sales is less ambiguous, as it does not appear to operate in the same manner as the other moderators, nor does it correlate highly with them ($r <$ .12 for both). However, the percentage of young adults and percentage of migration appear to operate in a parallel fashion and are highly correlated (r = .83). We attempted to tease apart their effects by regressing out the effect of the other moderator and using the residual as a moderator. The residual of young adults after regressing out the effect of migration no longer had a significant moderating effect; however, the residual of migration remained a significant moderator. These results may suggest that the moderating effect of migration rates is not due solely to its association with young adults or the mutual association of migration and young adults with some third variable. However, it is also possible that the difference between results from modeling the residuals of young adults and migration is due to scaling or range differences between the variables. A true test of the effect of the young adults versus the migration variable would require comparisons of areas in which there are both a high percentage of young adults and low migration with areas in which there is a high percentage of migration but proportionately few young adults. Because of the tight association of these variables, a comparison of this type would require a very large sample, and neighborhoods satisfying those requirements are probably unique and unrepresentative.

In other work, we have considered variables other than socioregional influences that could be involved in the moderating effect, but we have found none that accounted for the effect as strongly as did the three socioregional variables analyzed here (Dick, Viken, Kaprio & Rose, 2000). It is surprising that some of the more

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intuitively obvious explanations for the urban/rural effect were not supported by our data. Thus, it seemed plausible that twins in rural settings would have fewer peers available to them and would, therefore, spend more time with one another. That would result in their sharing more experiences, accounting for the increase in the importance of common environment found in rural settings. We investigated that possibility by comparing the mean difference between average social contact for co-twins in MZ and DZ pairs in urban and rural settings. We found no difference between urban and rural settings and no correlation between frequency of social contact and residency, suggesting that social contact did not contribute to the urban/rural interaction. We also considered whether the relevant difference was not time spent with the co-twin but how much the twins' peer groups overlapped. Once again, there was no MZ/DZ difference between the two settings, nor was there a correlation between common friends and residency. It is clear that individual, familial, and peer influences are not the sole critical factors involved in alcohol use; our analyses demonstrate the powerful effect of community-level factors. Socioregional variation deserves increased attention in future research, both to enhance our understanding of the development of trajectories of alcohol use and abuse and as a rich area for prevention and intervention efforts.

Many research groups have championed the importance of gene–environment interaction (e.g., Kendler & Eaves, 1986), and other methods previously have been proposed to study gene–environment interaction, such as the regression approach detailed by Rowe and Waldman (1993). This approach also allows for continuous environmental moderators to be incorporated into genetically informative analyses. However, biometrical modeling has increasingly become the analytic tool of choice among behavior geneticists because of the many advantages that maximum-likelihood methods offer, including flexibility in statistically representing theoretical models, the ability to compare the fit of competing models, and the ease with which these models are extended to the multivariate case. Thus, regression techniques have not been widely used. The methods we have described allow for better characterization of gene–environment interaction in biometrical twin models through the incorporation of more continuous and detailed measures of the environment. Our results suggest that extrafamilial environments can exert potent moderating effects on the expression of genetic and environmental influences on adolescent drinking patterns. Note that the current model assumes a linear relationship between the moderator and change in the importance of genetic and environmental effects. When the relation function is more complex, further elaborations would be necessary. Mx is very flexible in representing nonlinear relationships (Neale & McArdle, 2000), so adaptation to nonlinear patterns of gene–environment interaction is possible.

We conclude with a comment, and a caveat, on the nature of gene–environment interaction that our data demonstrate. We have shown that the magnitude of genetic influence (on drinking frequency) is moderated by specific measured attributes of rural and urban environments. Genetic variation is more influential in environments characterized by a greater percentage of young adults, higher migration rates, and proportionately greater alcohol sales. We have compared pairs of MZ and DZ twins who reside in environments varying in these attributes. However, we have not measured the genotypes of our twin subjects, nor have we manipulated the environments to which they were exposed. Our design is a naturalistic one, exploiting existing environmental variation between twin pairs to explore consequences for the expressed magnitude of genetic influences on their behavior. Such designs enjoy enhanced generalizability at the expense of some constraints. One critical constraint is that we cannot ignore the possibility that the environmental modulation we found arose from selection processes through which families of adolescent twins chose to reside in rural or urban settings; that possibility is commonly referred to as gene–environment correlation (Plomin, DeFries & Loehlin, 1977; Scarr & McCartney, 1983). However, were gene–environment correlations influential in our data, we would expect the drinking patterns of adolescents in urban and rural environments to differ. In fact, we found no significant difference in drinking patterns between urban and rural environments (Rose et al., 2001), suggesting that the differential impact of genes is not attributable to selection and allowing us to more confidently infer the presence of gene–environment interaction.

More powerful demonstrations of Gene × Environment interaction use experimental, not naturalistic, designs, exposing known genetic variation to manipulated changes in the environment. An illustration, with human subjects, is found in dietary overload studies (Bouchard et al., 1990), in which observed variation in weight gain and fat distribution between pairs of MZ twins was 2 to 5 times greater than the variation found within pairs. Emerging technology in molecular genetics now permits investigators to measure the response of measured genotypes to manipulated variation in the environments to which those genotypes are exposed. A persuasive example comes from studies showing that an apolipoprotein (ApoE4) polymorphism conditions individual response to dietary change: Subjects who carry the ApoE4 allele are significantly more sensitive to dietary interventions (Abbey, 1992). Soon, psychologists may engage in similar studies, contrasting individual differences in behavioral response to controlled environmental variations in subjects exhibiting known allelic variation. At that time, demonstrations of Gene × Environment interaction in human behavior will become commonplace. The illustration we report in this article suggests that powerful moderating effects of the environment can be made evident in analyses available now, while we await the promise of molecular genetics to be realized in routine genotyping of our subjects.

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